

Use case	Scenario
Data collection and analytics	Instead of sending back huge volumes of data to the cloud, deploy the analytics near the sensor site only, and pass the crux of analytics to the cloud for long-term trending and forecasting purposes.
Security	With the proliferation of mobiles and sensors, the chances of getting attacked also increase. Therefore, moving the security element closer to the originating source of the attack makes sense.
Compliance	This ranges from the locality of data to the ownership control – limiting data streaming back to the cloud makes compliance easier.
Network function virtualisation	This is the next generation of network infrastructure elasticity, where the telcos are increasingly trying to run their networking on top of NFV to reduce the cost, instead of being centrally managed.
Real-time	The advent of AR/VR, self-driven cars, telemedicine, smart cities and so on, adds flavour to cloud-edge computing.
Immersive	A new breed of applications includes AR/VR, 4K video, and 360° imaging for verticals like healthcare.
Network efficiency	Not all the applications need to be run centrally. Facial recognition or number plate recognition fit well into the edge itself.
Self-contained and autonomous site operations	A stable network is one of the challenges of edge for hostile sites like mining, power generation, etc. Hence, running wholistic apps in the edge is beneficial for these types of deployments.

Table 1: Cloud-edge use cases [Reference: <https://www.openstack.org/use-cases/edge-computing/cloud-edge-computing-beyond-the-data-center/>]

Tenets	Role
Workload analysis and modelling	1. Understand the IO pattern (IO size, % read, % write, concurrency). 2. Generate a signature (DNA) from the IO pattern. 3. Model the IO pattern signature vs the load graph. 4. Predict the load from a new IO pattern.
Application optimisation	1. Maintain performance guarantee as the main goal. 2. Elasticity and remediation are the key components. 3. Load balancing is maintained for elasticity. 4. Autoscaling is a must for pay-as-you-go.
Infrastructure optimisation	1. Software Defined Infrastructure (SDI) is to abstract out the resources from the underlying most important aspects of the infrastructure. 2. Resource placements are dynamically programmable in SDI. 3. It ensures resource flexibility in the multi-provider environment. 4. It optimises the utilisation of the infrastructure resource and the underlying network.

Table 2: Resource provision tenets

IoT adds another dimension in this context. IoT devices are big storehouses of data. They are designed to operate in hostile environments such as unstable networks, with constraints of memory and storage, and so on. Hence, to provide contextual and personalised recommendations, the overall system architecture needs to process huge amounts of data and keep on transferring the digest from the edge towards cloud. In such a scenario, maintaining the integrity of the continuum resource availability from edge to cloud is of paramount importance.

To understand the problem at hand, one must look at other aspects too like the standardisation of the IoT ecosystem. To facilitate the convergence of networking and IT, the Mobile Edge Computing (MEC) Working Committee has proposed a model in which compute (in the cloud)